Difference of means - Practice

MATH 213

Group 1

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the “age at first marriage” variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	$^{5}/_{\sqrt{1395}}$ = 0.134
SE for mean age at first marriage for 2006	$^{6.1}/_{\sqrt{1161}}$ = 0.179
SE for difference in means (2006 mean – 1972 mean)	$\sqrt{(0.134^2 + 0.179^2)}$ = 0.224

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

0.224 * 2 = 0.448 23.3 - 22.1 = 1.2
1.2 - 0.448 = 0.752 1.2 + 0.448 = 1.648 (0.752, 1.648)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval: CI: (0.7616, 1.6384)

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the true difference in the mean age between people being married in 2006 for the first time and people in 1972 being married for the first time is between 0.752 and 1.648. That is, the average person in 2006 was married for the first time between 0.752 and 1.648 years later in life than the average person in 1972.

Group 2

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the “age at first marriage” variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	0.134
SE for mean age at first marriage for 2006	0.179
SE for difference in means (2006 mean – 1972 mean)	0.224

Can use this space for work.
1972_SE = 5 / sqrt(1395) = ~0.134

2006_SE = 6.1/sqrt(1161) = 0.179

SE_diff = sqrt(0.134^2 + 0.179^2)

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

95_ci = 1.2 +/- 2(0.224) = (0.752, 1.648)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval: (0.7616, 1.6384)

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the true difference of the population's average age at first marriage when comparing 2006 and 1972 is between 0.7616 and 1.6384 years

Group 3

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the “age at first marriage” variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	5/square root 1395 = 0.134
SE for mean age at first marriage for 2006	6.1/square root 1161 = 0.179
SE for difference in means (2006 mean – 1972 mean)	Square root (0.134)^2 + (0.179)^2 = 0.224

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

Diff in mean = 23.3 - 22.1 = 1.2

CIc= 1.2 +- 2 * 0.224 = (0.75,1.65)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval:

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that amoung US adults who have been married, the average age of the first marriage in 2006

Group 4

The General Social Survey, for many years, tracked the variable "age at first marriage" on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question "How old were you when you first married?"

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the "age at first marriage" variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	0.134
SE for mean age at first marriage for 2006	0.179
SE for difference in means (2006 mean – 1972 mean)	0.224

Can use this space for work.

SE_1972 = 5/sqrt(1395) = 0.134

SE_2006 = 6.1/sqrt(1161) = 0.179

SE_diff = geometric average of 0.134 and 0.179 = 0.224

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

1.2 - 2 * 0.224 = 0.752

1.2 + 2 * 0.224 = 1.648

95%_CI = (0.752, 1.648)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.
Rossman/Chance confidence interval:
The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!
We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the difference in population mean will appear in our interval for the average age of first marriage in the US between 1972 and 2006 is between 0.752 and 1.648 years (2006-1972) according to the data collected in the General Social Survey.

Group 5

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the “age at first marriage” variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	.133
SE for mean age at first marriage for 2006	.179
SE for difference in means (2006 mean – 1972 mean)	.221

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

1.2+2*.221=1.642

1.2-2*.221=.758

(0.758, 1.642)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate “scenario” before beginning.

Rossman/Chance confidence interval:

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the the average age for getting married in 1972 on average is .758 to 1.6 years less in 2006.

Group 6

The General Social Survey, for many years, tracked the variable "age at first marriage" on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question "How old were you when you first married?"

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the "age at first marriage" variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	0.13
SE for mean age at first marriage for 2006	0.18
SE for difference in means (2006 mean – 1972 mean)	0.22

Can use this space for work.

SE for mean age at first marriage 1972: 5 / sqrt(1395) = 0.13

SE for mean age at first marriage 2006: 6.1 / sqrt(1161) = 0.18

SE for difference in means: sqrt($0.13^2 + 0.18^2$) = 0.22

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

→ 1.2 +- (2 * 0.22)

95% CI: (0.76, 1.64)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval: (0.7616, 1.6384)

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.) We are 95% confident that the difference in population's mean age of first marriage between 2006 and 1972 is between 0.76 and 1.64.

95 times out of 100 (on average), the 95% confidence interval will contain the population's difference in means.

Group 7

The General Social Survey, for many years, tracked the variable "age at first marriage" on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question "How old were you when you first married?"

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the "age at first marriage" variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	5/sqrt(1395) = 0.134
SE for mean age at first marriage for 2006	6.1/sqrt(1161) = 0.179
SE for difference in means (2006 mean – 1972 mean)	0.224

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

1.2 +- 0.224 = (0.752, 1.648)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval:

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the true difference between means of peoples age of first marriage in 1972 and 2006 is between 0.752 and 1.648 years

Group 8

The General Social Survey, for many years, tracked the variable "age at first marriage" on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question "How old were you when you first married?"

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the "age at first marriage" variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	5/sqroot(1395) = .133
SE for mean age at first marriage for 2006	6.1/sqroot(1161) = .179
SE for difference in means (2006 mean – 1972 mean)	sqroot((.133)^2 + (.179)^2) =.223

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

-1.2 -2(.223) = -1.646

-1.2 +2(.223) = -.754

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval: -1.634 - -0.761

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confidence that the mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who have ever been married is between -1.646 and .754.

Group 9

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the “age at first marriage” variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	0.133
SE for mean age at first marriage for 2006	0.179
SE for difference in means (2006 mean – 1972 mean)	0.046

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

23.3-22.1 = 1.2 pop parameter

0.133^2 + 0.179^2 =square root (0.017 + 0.032)

1.2 +- 2*0.221 = (1.642, 0.758)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate “scenario” before beginning.

Rossman/Chance confidence interval:

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.) We are 95% confident that the difference in population average age of first marriage in the US between 1972 and 2006 are between 0.758 and 1.642.

Group 10

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

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Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	5 / sqrt(1395) = 0.134
SE for mean age at first marriage for 2006	6.1 / sqrt(1161) = 0.179
SE for difference in means (2006 mean – 1972 mean)	sqrt(0.134^2 + 0.179^2) = 0.224

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

23.3 - 22.1 = 1.2

1.2 +- 2 * 0.224 = (0.75, 1.65)

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval: (0.762, 1.638)

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the difference in the mean age of first marriages in 2006 was between 0.75 and 1.65 years higher than the mean ages of first marriages in 1972.

Group 11

The General Social Survey, for many years, tracked the variable “age at first marriage” on the survey. Remember that the GSS surveys a representative sample of US residents each year. In the survey sample from the year 1972, for example, there were 1395 people who were currently married, widowed, divorced, or separated. These people responded to the question “How old were you when you first married?”

For some reason, the GSS has stopped asking about age at first marriage in their surveys. The last year when the survey included this question was 2006. The following table summarizes the “age at first marriage” variable for these two years.

Survey year	Sample size	Mean age at first marriage	Standard deviation in age at first marriage
1972	1395	22.1	5.0
2006	1161	23.3	6.1

1. Use the sample data to calculate the following estimated standard errors. Show some of your work.

SE for mean age at first marriage for 1972	5.0/sqrt(1395)=0.13386988815
SE for mean age at first marriage for 2006	6.1/sqrt(1161)=0.17902501762
SE for difference in means (2006 mean – 1972 mean)	sqrt(0.13386988815^2+0.17902501762 ^2)=0.22354217473

Can use this space for work.

2. Use the appropriate standard error and appropriate statistic to create a 95% confidence interval for the difference in mean age at first marriage among US adults in 2006 who had ever been married and among US adults in 1972 who had ever been married. Show some work.

22.1-23.3= -1.2

1.2+ or - 2* 0.22354217473

3. Check your work using the Rossman/Chance TBI applet - link in blackboard. Make sure to select the appropriate "scenario" before beginning.

Rossman/Chance confidence interval: 0.7616, 1.6384

The applet interval should be very close to what you found in part b. If it is not, check your work to figure out what went wrong!

We can also check this in R. See our R Cheat Sheet for some new code for calculating confidence intervals. (Slides near the end.)

4. Write a sentence of interpretation for the 95% confidence interval for difference in means. (You may wish to look up the Difference of Proportions question we did in class on Friday (the sentence with the fill-in-the-blanks at the end) to get an idea for how to write such a sentence.)

We are 95% confident that the difference in average age is between 0.76 and 1.6 years